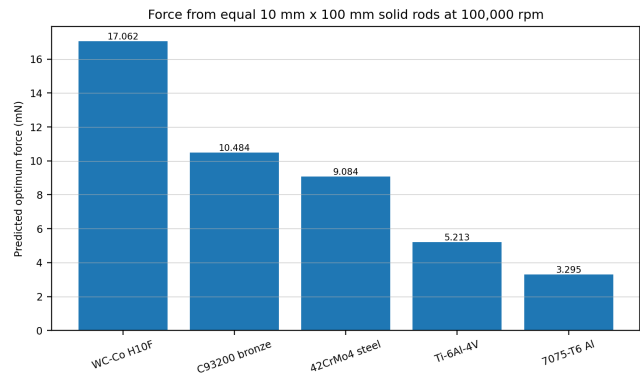
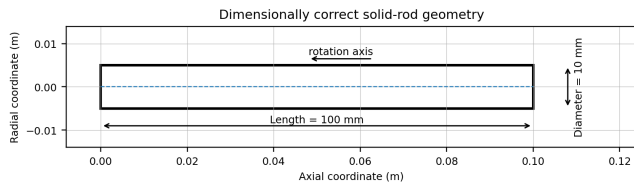


Rod Material Comparison

Force, density scaling, centrifugal stress, tensile-strength margins, and rotor-dynamic constraints for a 10 mm x 100 mm solid rod at 100,000 rpm



Title figures. The rod drawing uses the exact 10 mm diameter and 100 mm length. The force comparison is calculated from the stated material densities, the locked K-Field coefficient, the optimum Earth-CMB-relative orientation, and 100,000 rpm.

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This paper is a standalone educational and archival reference. It defines the complete geometry, Cell3 coordinate lock, K-Field force calculation, established rotating-solid stress bounds, representative material grades, tensile and yield margins, strength-limited speed estimates, and rotor-dynamic constraints. Numerical figures and tables are generated from the machine-readable companion ledger.

Abstract

A uniform solid rod 10 mm in diameter and 100 mm long has a volume of $7.853981633974 \times 10^{-6} \text{ m}^3$. At 100,000 rpm its angular speed is 10471.975512 rad/s and its outer-rim velocity is 52.359878 m/s. Under the current optimum-orientation K-Field law, every material has the same specific force, 0.149822768 N/kg. Total force is therefore proportional to density.

The selected grades are Hyperion H10F WC-Co cemented carbide, Kaiser 7075-T6/T651 aluminum, annealed Ti-6Al-4V Grade 5 bar, quenched-and-tempered 42CrMo4/AISI 4140 steel, and continuous-cast C93200 bronze. Their predicted forces are approximately 17.062 mN, 3.295 mN, 5.213 mN, 9.084 mN, and 10.484 mN, respectively.

A deliberately conservative centrifugal-stress bound treats all material as if it were concentrated at the outer radius: $\sigma = \rho v^2$. The resulting stresses range from 7.68 MPa for aluminum to 39.75 MPa for H10F carbide. Conservative yield safety factors are approximately 65.5 for aluminum, 71.0 for titanium, 42.5 for steel, and 5.65 for bronze. The carbide TRS-to-stress comparison is approximately 113, but ASTM B406 states that TRS is flaw-sensitive and is not a design value.

The bulk centrifugal stress is not the principal design limitation at the stated speed. The 100 mm length-to-diameter ratio is 10, so balance, bearing support, bending critical speeds, attachment stresses, material anisotropy, defect tolerance, and fragment containment govern practical operation.

Central result

For equal geometry and equal rpm, density increases K-Field force and centrifugal stress in the same linear proportion. Material selection is therefore not determined by density alone. Force per rotating mass and force per stored spin energy are identical across the material set; safety is controlled primarily by strength-to-density ratio, flaw tolerance, rotor dynamics, and support design.

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2	Rod geometry and rotational state
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8	Centrifugal stress models
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1. Scope, selected grades, and coordinate lock

The word aluminum, titanium, steel, bronze, or tungsten carbide does not identify a unique engineering material. Strength depends on alloy, heat treatment, product form, section size, grain structure, and manufacturing quality. This comparison therefore uses specific representative grades rather than generic family averages.

Material family	Selected grade or condition	Reason for selection
WC-Co H10F	WC-Co cemented carbide H10F	Typical transverse rupture strength (TRS); not a design tensile value
7075-T6 Al	7075-T6/T651 aluminum rod and bar	Typical 0.2% yield and ultimate tensile strengths
Ti-6Al-4V	Ti-6Al-4V Grade 5 annealed bar	Minimum bar yield and ultimate tensile strengths for diameter below 2 in
42CrMo4 steel	42CrMo4 / AISI 4140 quenched and tempered steel	Minimum yield and lower-bound UTS for $d \leq 16$ mm
C93200 bronze	C93200 continuous-cast bearing bronze	Minimum 0.5% extension yield and ultimate tensile strengths

The coordinate frame is fixed ICRS/J2000 Cartesian. The Earth-CMB unit vector $\hat{c}$, Earth translation velocity V_E , Cell3 tensor G_{phase} , transformed response $\hat{g}$, optimum spin axis $\hat{n}_{\text{opt}}$, and lateral force direction $\hat{F}$ are common to every material. Only the mass-dependent mechanical quantities change.

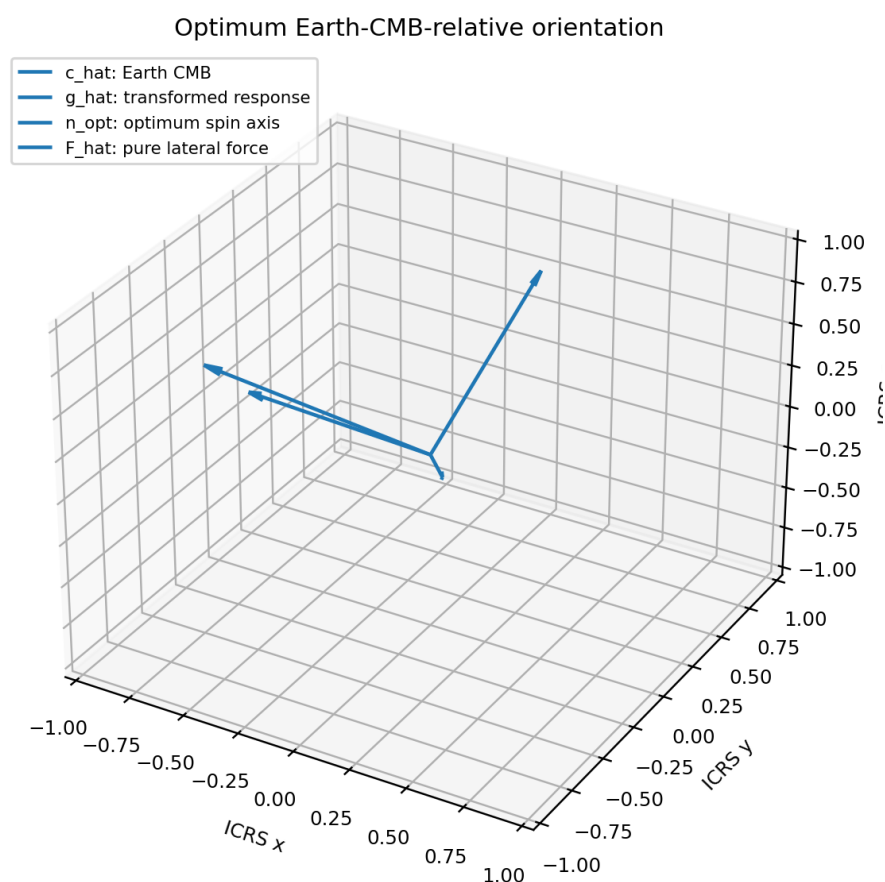


Figure 1. Exact optimum vector construction. The spin axis is perpendicular to $\hat{g}$, and the force is perpendicular to both the Earth-CMB vector and $\hat{g}$.

2. Rod geometry and rotational state

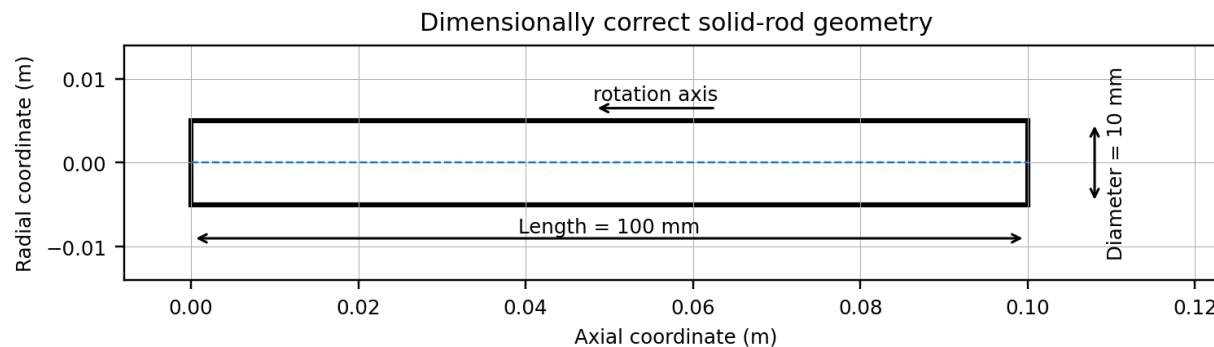


Figure 2. Dimensionally correct common rod envelope. Every material occupies the same cylindrical volume and rotates about the longitudinal centerline.

Quantity	Value	Unit
Diameter D	0.010000000	m
Radius R	0.005000000	m
Length L	0.100000000	m
Volume $\pi R^2 L$	7.853981633974484e-06	m ³
Cross-sectional area	7.853981633974483e-05	m ²
Bending second moment $\pi R^4/4$	4.908738521234052e-10	m ⁴
Polar second moment $\pi R^4/2$	9.817477042468103e-10	m ⁴
Spin speed	100000.000	rpm
Spin frequency	1666.666666667	Hz
Angular speed ω	10471.975511965977	rad/s
Outer-rim speed	52.359877559830	m/s

At this speed the rim velocity is only 52.36 m/s. The high numerical rpm follows from the small 5 mm radius.

3. Authoritative Cell3 base cell and plane families

The K-Field force vector is evaluated in the same fixed ICRS frame as the Cell3 basis. The centered primitive cell is generated from the authoritative basis vectors a , b , and c . The device spin axis is a separate vector in this common frame.

```
a = (0.020381438516, -0.003416414496, -0.002332965916)
b = (0.002198745968, 0.012902170258, -0.034066501536)
c = (-0.000630379486, -0.024257722018, -0.042924354572)
B_Cell3 = [a b c]
```

Authoritative centered Cell3: six faces and all 27 half-step nodes

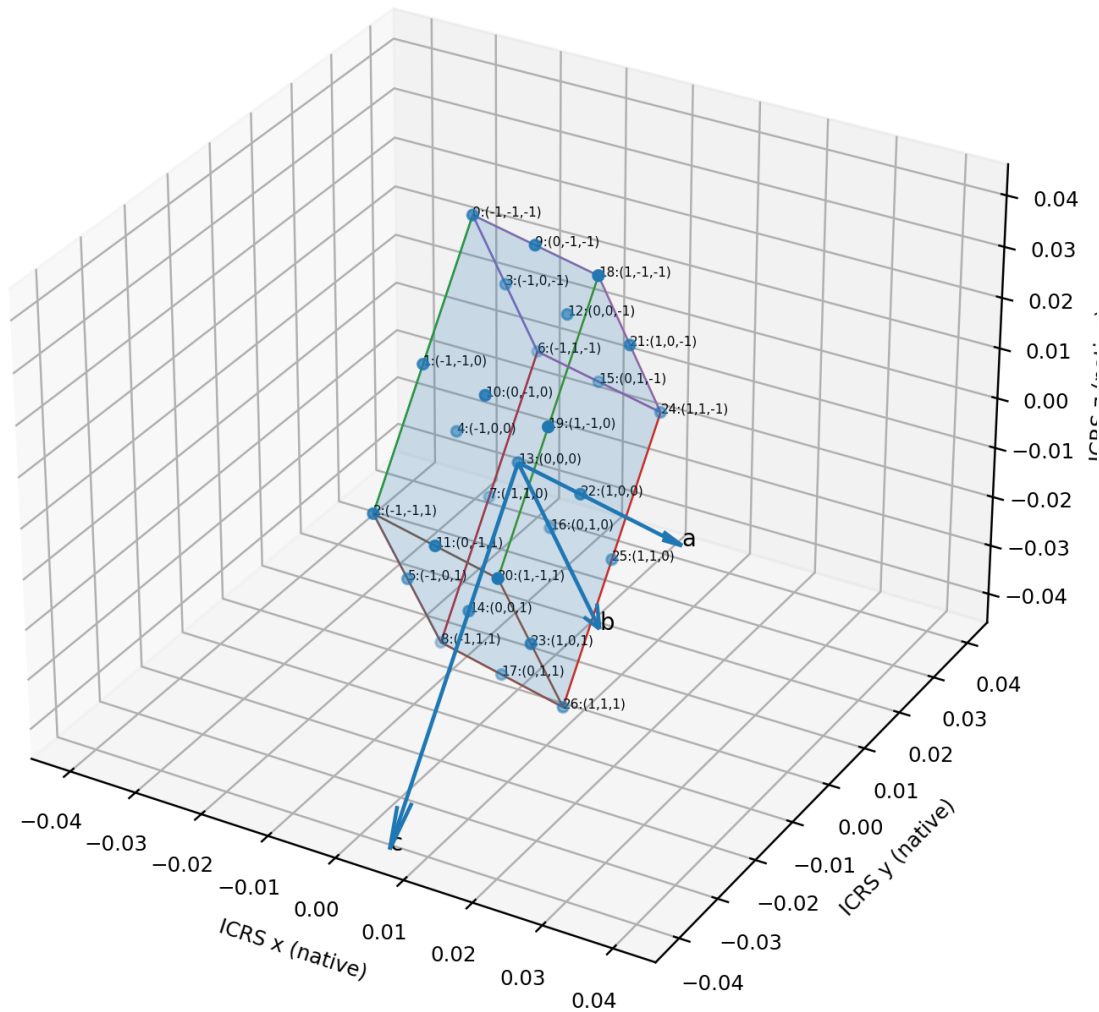


Figure 3. The complete centered Cell3, including six faces, twelve edges, all 27 half-step address nodes, integer labels, and basis vectors.

Cell3 central plane families ab, ac, and bc

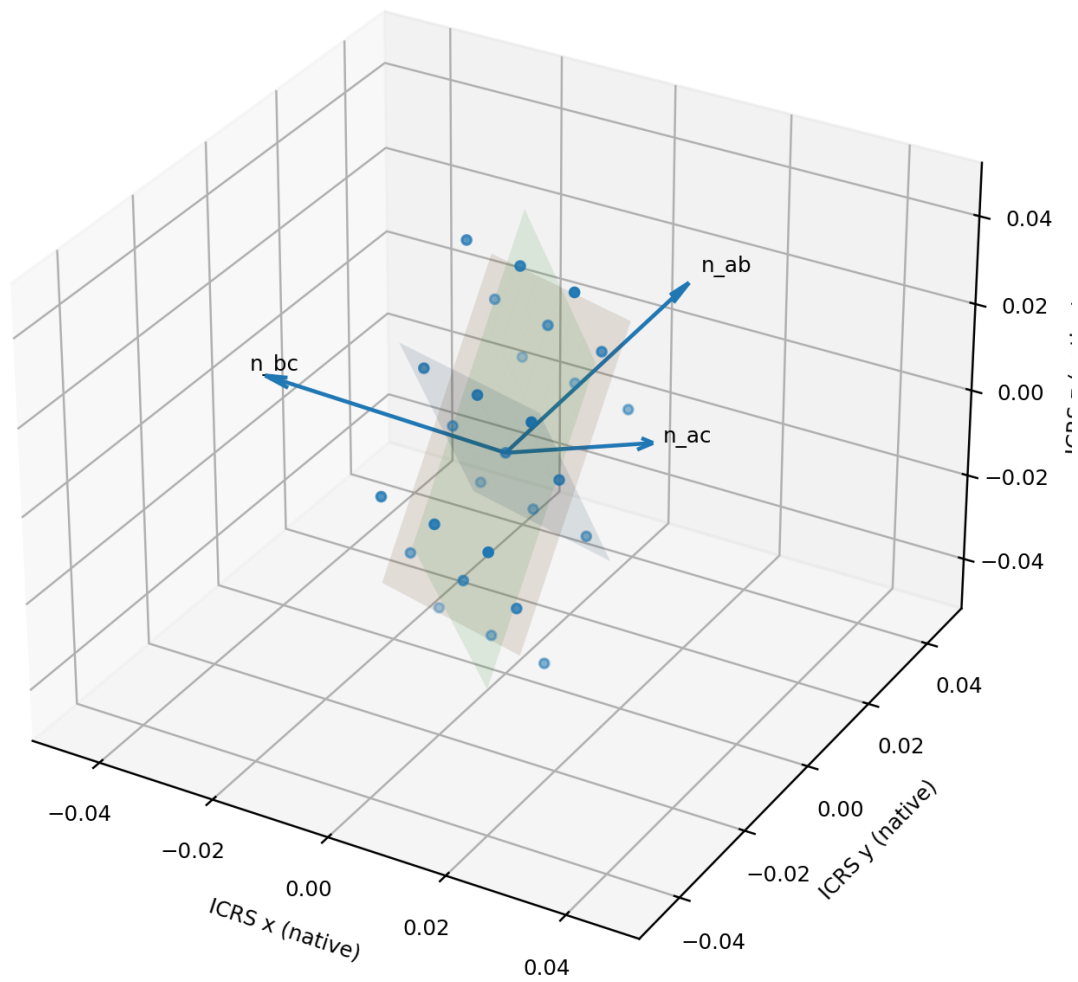


Figure 4. Central ab, ac, and bc plane families, generated from their exact spanning vectors and fixed ICRS normals.

Family	Area native ²	Included angle	Unit normal ICRS
A_ab	7.547217896293e-04	83.933483014 deg	(0.1940922008, 0.9131771688, 0.3583792348)
A_ac	1.011255757668e-03	80.448130021 deg	(0.0890525941, 0.8665767693, -0.4910339483)
A_bc	1.385784431744e-03	50.363231813 deg	(-0.9959651903, 0.0836021625, -0.0326192906)

4. K-Field force law and optimum orientation

For a complete axisymmetric rotor, the integrated force law is

$$\mathbf{F}_K = (\kappa_F m \omega / 2) [\hat{n} \text{ cross } (G_{\text{phase}} \mathbf{V}_E)].$$

Define $\mathbf{g} = G_{\text{phase}} \mathbf{V}_E$ and $q_g = \kappa_F |\mathbf{g}|/2$. In the optimum orientation, $\hat{n}$ is perpendicular to $\mathbf{g}$, so

$$|\mathbf{F}_K|_{\text{max}} = m q_g \omega$$

$$|\mathbf{F}_K|_{\text{max}}/m = q_g \omega.$$

Quantity	Value	Unit or role
c_hat	(-0.970770855836, 0.207348009111, -0.120874929482)	unitless ICRS
V_E	(-359003.199066, 76679.886034, -44701.060101)	m/s
g	(-880221.981635, -143911.436562, 266335.026949)	m/s effective response
g	930825.217270519235	m/s
kappa_F	3.0740508997020345e-11	coupling
q_g	1.430702048307891e-05	N kg ⁻¹ per rad s ⁻¹
Specific force at 100k rpm	0.149822768148	N/kg

The material density does not change the acceleration predicted for a free rotor. Density changes mass, total force, centrifugal stress, inertia, and stored energy.

5. Material property basis and source control

The force calculation requires density. The safety calculation additionally requires a strength basis. Ductile materials are compared against both yield and ultimate tensile strengths. Cemented carbide has negligible ductility and is reported using transverse rupture strength; that number requires separate qualification.

Material	Density kg/m ³	Yield MPa	UTS MPa	TRS MPa	Elastic modulus GPa	Sources
WC-Co H10F	14500	-	-	4500.0	-	S1,S6
7075-T6 Al	2800	503.0	572.0	-	71.00	S2
Ti-6Al-4V	4430	862.0	931.0	-	110.00	S3,S4
42CrMo4 steel	7720	900.0	1100.0	-	210.00	S5
C93200 bronze	8910	137.9	241.3	-	99.97	S7

For 42CrMo4, the selected minimum properties apply to quenched-and-tempered round sections no larger than 16 mm. For C93200, continuous-cast minimum properties are used. For aluminum, the selected values are Kaiser T6/T651 rod-and-bar typical values. For titanium, the selected bar minimums reproduce the declared comparison basis; TIMET data provide a conservative cross-check. All properties are room-temperature values unless noted.

6. Mass and force calculation

Every rod has the same volume. Mass is therefore directly proportional to density:

$$m = \rho \pi R^2 L.$$

The optimum K-Field force is

$$F_K = m q_g \omega = \rho \pi R^2 L q_g \omega.$$

Material	Mass (g)	Force (mN)	Equivalent static gf	Specific force (N/kg)
WC-Co H10F	113.882734	17.062226	1.739863	0.149822768148
C93200 bronze	69.978976	10.484444	1.069116	0.149822768148
42CrMo4 steel	60.632738	9.084165	0.926327	0.149822768148
Ti-6Al-4V	34.793139	5.212804	0.531558	0.149822768148
7075-T6 Al	21.991149	3.294775	0.335974	0.149822768148

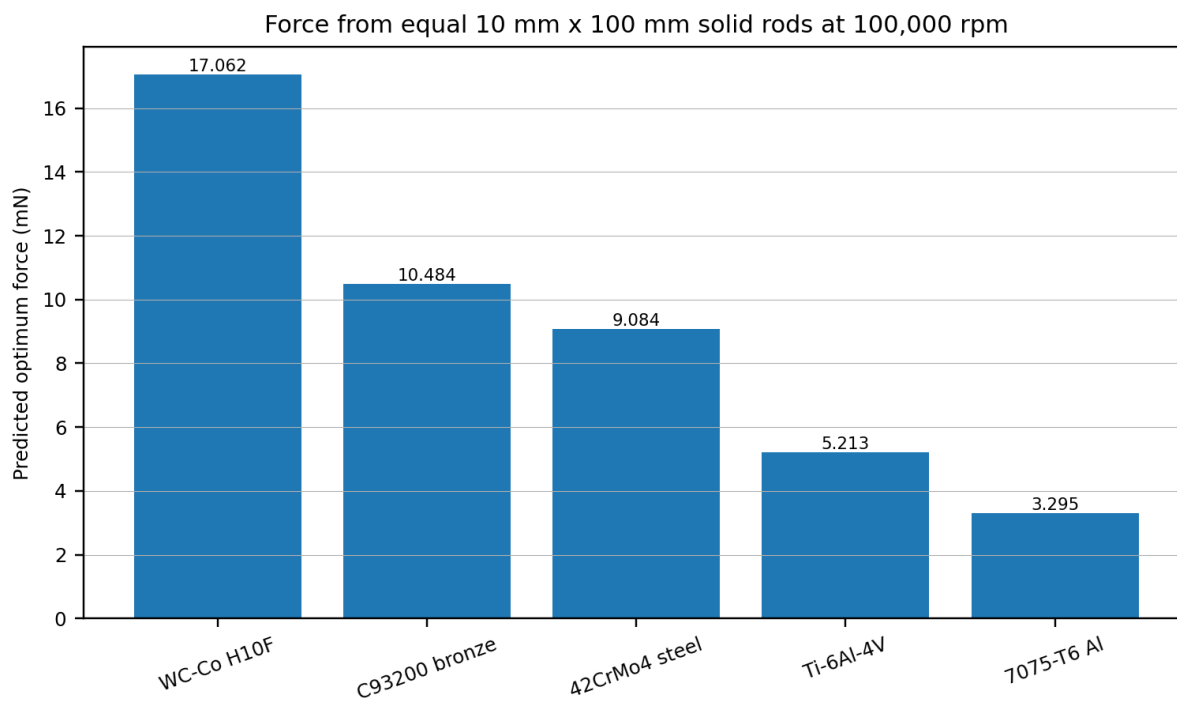


Figure 5. Total predicted force for the equal-volume rods. The density ordering determines the force ordering.

7. Force ranking and density scaling

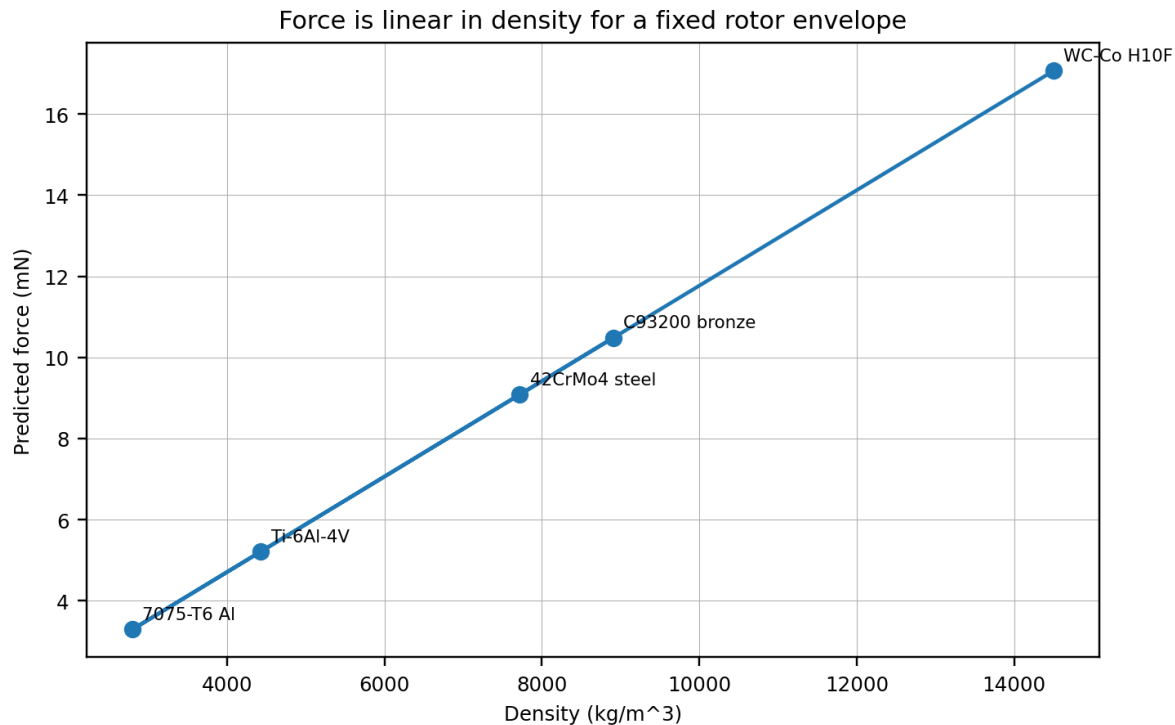


Figure 6. The five calculated points lie on the exact linear density-force relation for the common geometry and rpm.

The fixed-envelope force ranking is

WC-Co H10F > C93200 bronze > 42CrMo4 steel > Ti-6Al-4V > 7075-T6 aluminum.

Material	Force relative to aluminum	Interpretation
WC-Co H10F	5.178571	Equal to density ratio
C93200 bronze	3.182143	Equal to density ratio
42CrMo4 steel	2.757143	Equal to density ratio
Ti-6Al-4V	1.582143	Equal to density ratio
7075-T6 Al	1.000000	Equal to density ratio

H10F carbide produces approximately 5.18 times the aluminum force; bronze 3.18 times; steel 2.76 times; and titanium 1.58 times. These gains apply to force per fixed geometric envelope. Force per unit rotating mass remains identical.

8. Centrifugal stress models

Two stress estimates are retained. The first deliberately overbounds the solid rod by treating all mass as if it were concentrated in a thin ring at the outer radius:

$$\sigma_{\text{ring}} = \rho v_{\text{rim}}^2 = \rho \omega^2 R^2.$$

For an ideal isotropic solid disk in plane stress, the maximum center stress is

$$\sigma_{\text{center}} = ((3 + \nu)/8) \rho \omega^2 R^2.$$

Because a stable isotropic material has $\nu \leq 0.5$,

$$\sigma_{\text{center}} \leq (7/16) \rho \omega^2 R^2 = 0.4375 \sigma_{\text{ring}}.$$

The thin-ring result is used for the primary safety factors because it is more severe and does not require a Poisson-ratio assumption.

Material	Thin-ring stress MPa	Ideal-solid upper MPa	Strength basis MPa
WC-Co H10F	39.752573	17.391751	4500.0
7075-T6 Al	7.676359	3.358407	503.0

Material	Thin-ring stress MPa	Ideal-solid upper MPa	Strength basis MPa
Ti-6Al-4V	12.145097	5.313480	862.0
42CrMo4 steel	21.164818	9.259608	900.0
C93200 bronze	24.427271	10.686931	137.9

Centrifugal self-stress compared with strength basis

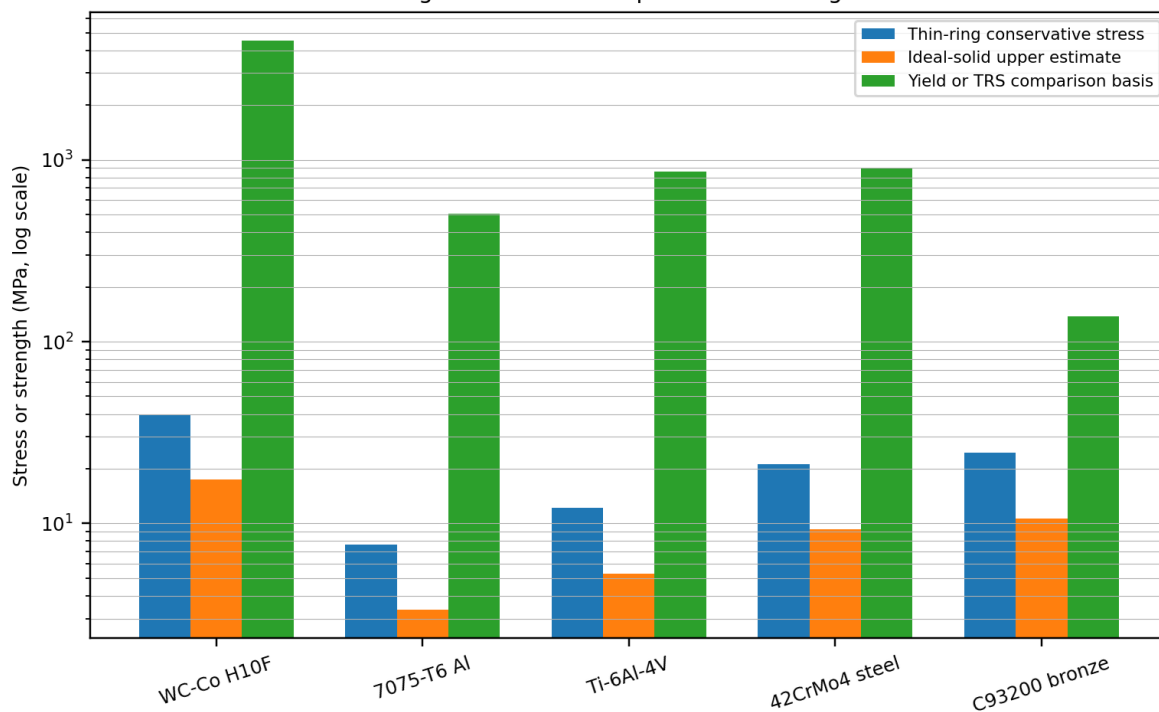


Figure 7. The logarithmic scale shows the large separation between bulk centrifugal self-stress and the selected strength basis at 100,000 rpm.

9. Tensile, yield, and TRS safety factors

For ductile metals, yielding is the preferred first limit because permanent deformation can destroy balance before tensile fracture. Ultimate tensile strength is also listed to preserve the complete fracture comparison. For carbide, the displayed number is a TRS-to-stress comparison rather than a certified safety factor.

Material	Ring stress MPa	Yield SF	UTS SF	TRS/stress	Ideal-solid yield or TRS ratio
WC-Co H10F	39.7526	-	-	113.20	258.74
7075-T6 Al	7.6764	65.53	74.51	-	149.77
Ti-6Al-4V	12.1451	70.98	76.66	-	162.23
42CrMo4 steel	21.1648	42.52	51.97	-	97.20
C93200 bronze	24.4273	5.65	9.88	-	12.90

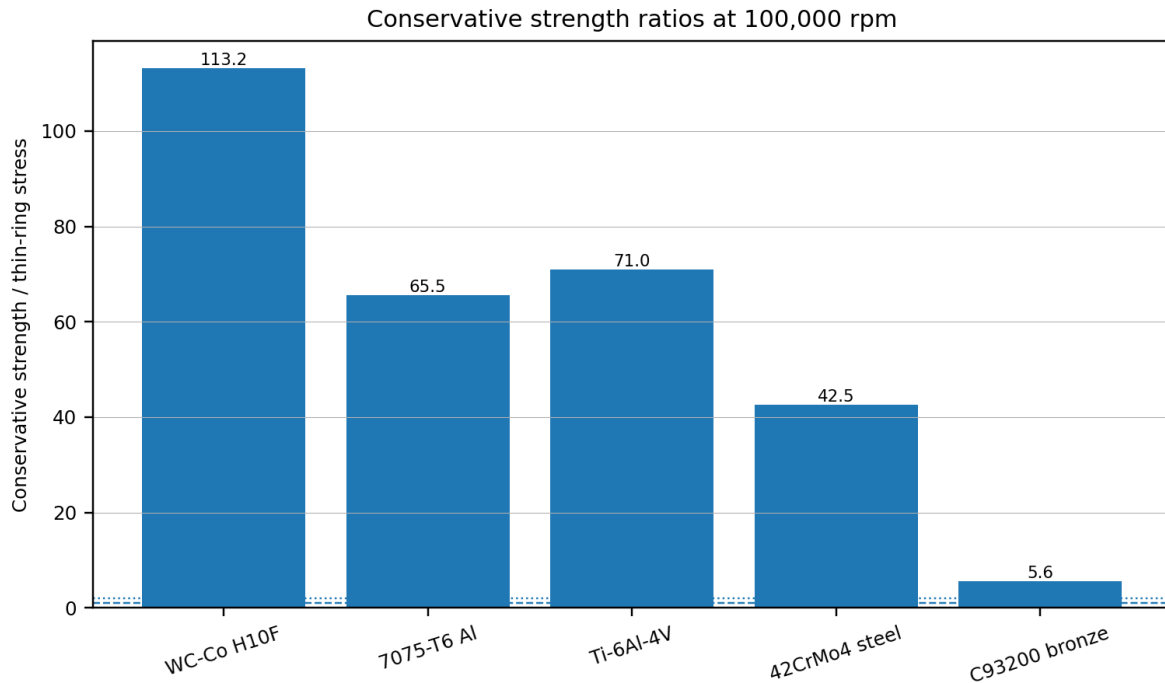


Figure 8. Conservative strength ratios based on the thin-ring stress. Bronze has the lowest ductile-material yield margin, approximately 5.65.

The conservative values reproduce the central comparison: H10F carbide approximately 113 by TRS ratio; aluminum yield 65.5 and UTS 74.5; titanium yield 71.0 and UTS 76.7; steel yield 42.5 and minimum-UTS 52.0; bronze yield 5.65 and UTS 9.88.

10. Strength-limited speed envelopes

Under the thin-ring bound, stress increases with the square of rpm. Solving $\sigma = S/N$ for rpm gives

$$\text{RPM}_{\text{allow}} = [60/(2 \pi R)] \sqrt{S/(N \rho)},$$

where S is yield strength for ductile materials or TRS for the carbide comparison, and N is the selected strength ratio.

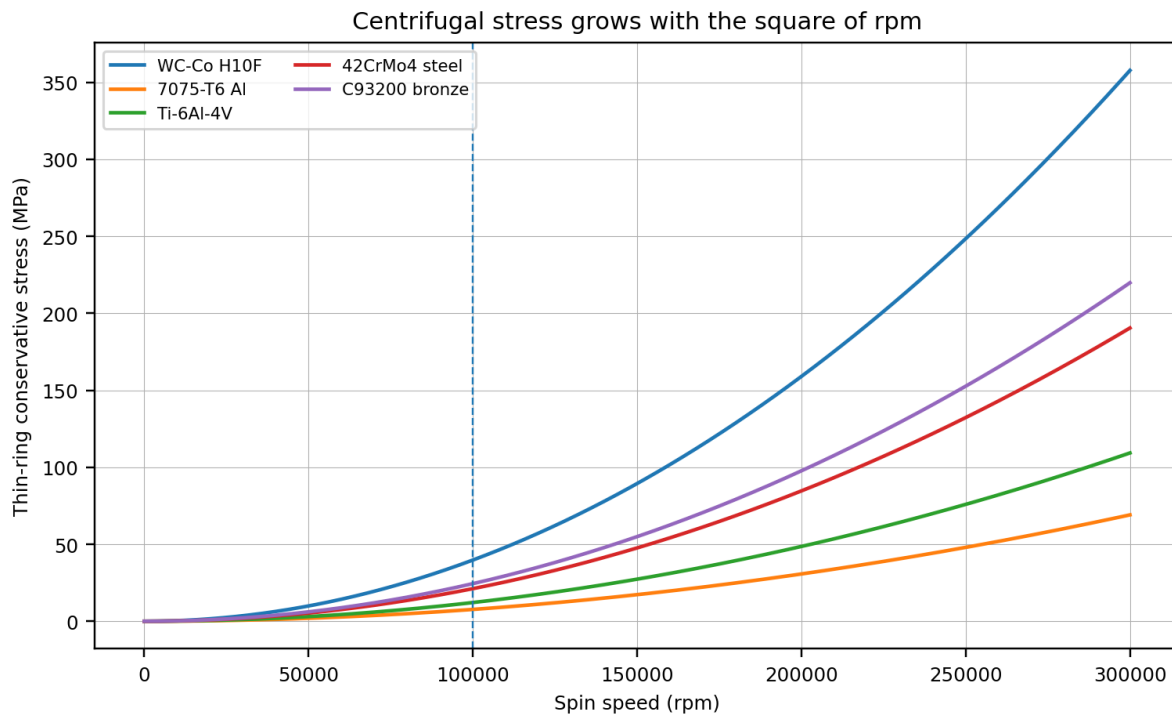


Figure 9. Conservative centrifugal stress versus rpm. All curves pass through zero and grow quadratically.

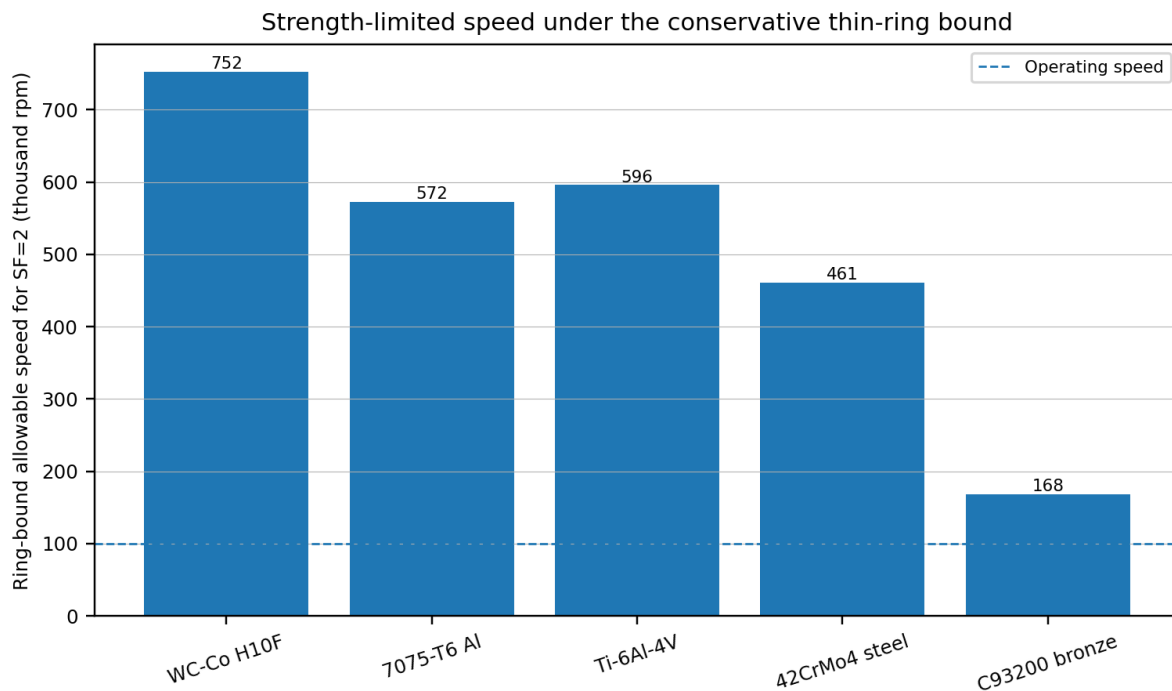


Figure 10. Thin-ring strength-limited speeds for a ratio of 2. Bronze is the most restrictive at approximately 168,000 rpm. These are material-stress envelopes only; rotor dynamics can impose a lower limit.

Material	Allowable rpm SF=1	Allowable rpm SF=2	Allowable rpm SF=3	Basis
WC-Co H10F	1,063,956	752,330	614,275	TRS comparison
7075-T6 Al	809,480	572,389	467,354	yield
Ti-6Al-4V	842,467	595,714	486,399	yield
42CrMo4 steel	652,100	461,104	376,490	yield
C93200 bronze	237,599	168,008	137,178	yield

11. Rotational inertia, angular momentum, and stored energy

The solid-cylinder polar inertia, angular momentum, and stored energy are

$$I = (1/2) m R^2$$

$$H = I \omega$$

$$E = (1/2) I \omega^2.$$

Material	Polar inertia kg m ²	Angular momentum kg m ² /s	Stored energy J	Force mN
WC-Co H10F	1.423534171158e-06	1.490721498081e-02	78.053995	17.062226
C93200 bronze	8.747372044839e-07	9.160226584761e-03	47.962834	10.484444
42CrMo4 steel	7.579092276785e-07	7.936806872543e-03	41.557024	9.084165
Ti-6Al-4V	4.349142329813e-07	4.554411197586e-03	23.846841	5.212804
7075-T6 Al	2.748893571891e-07	2.878634616984e-03	15.072496	3.294775

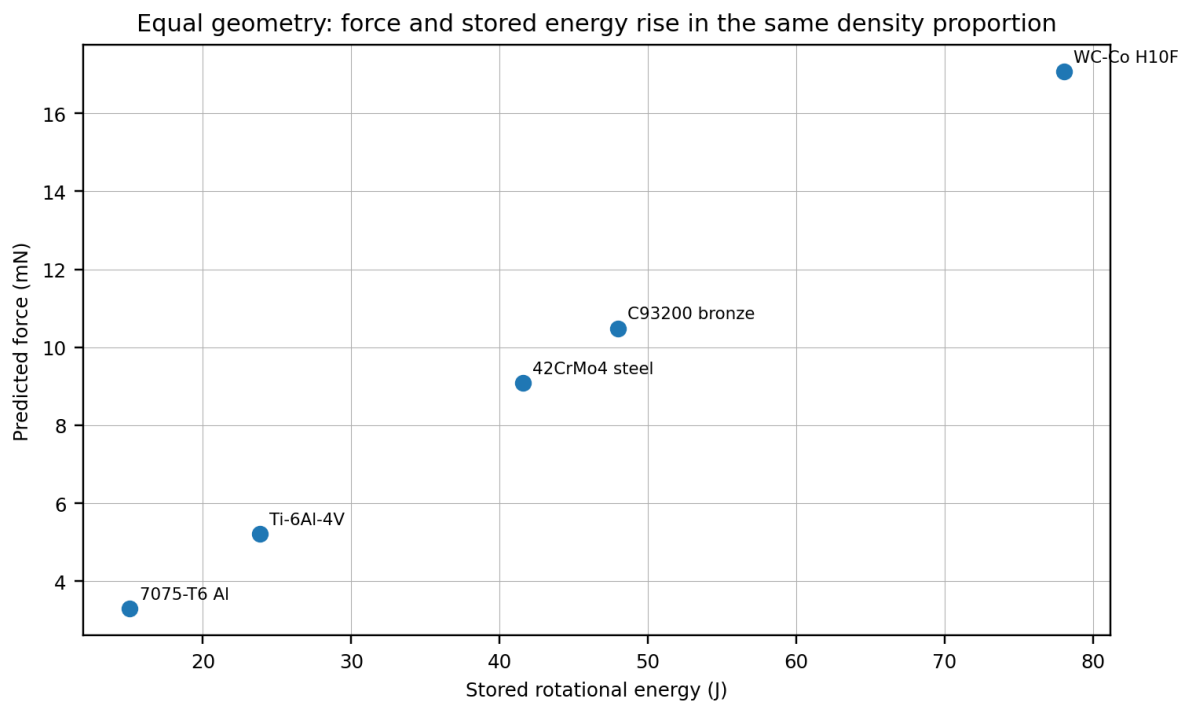


Figure 11. At common geometry and rpm, force and stored energy both scale with mass and therefore with density. The graph is linear.

12. Density-independent performance ratios

Several ratios simplify exactly for equal geometry and rpm.

$$F/m = q_g \omega.$$

This is identical for every material. A denser rod produces more force only because it contains more rotating mass.

$$F/E = [m q_g \omega] / [(1/4)m R^2 \omega^2]$$

$$= 4 q_g / (R^2 \omega).$$

For the present rod, $F/E = 2.185951709567e-04$ N/J for every material. Tungsten carbide's greater force requires proportionally greater stored rotational energy.

$$F/\sigma_{ring} = [\rho V q_g \omega] / [\rho \omega^2 R^2]$$

$$= V q_g / (\omega R^2).$$

Density also cancels from force divided by conservative centrifugal stress. Therefore a high-density material does not obtain a force advantage without an equal increase in self-stress. The decisive safety metric becomes strength divided by density.

These invariants separate three design questions: density selects force per package volume; the K-Field law fixes force per rotating mass; and strength-to-density plus rotor dynamics select the safe operating envelope.

13. Tungsten-carbide fracture qualification

H10F is a cemented WC-Co composite, not monolithic tungsten carbide. The selected grade contains approximately 89.5% WC and 10.5% binder by weight. Hyperion lists a typical density of 14.5 g/cm³, typical TRS of 4500 MPa, and compressive strength of 6300 MPa.

The approximately 113 TRS-to-ring-stress ratio must not be interpreted as an ordinary ductile-metal safety factor. ASTM B406 states that cemented-carbide TRS is not a design value. Fracture is controlled statistically by surface and internal flaws, specimen volume, surface preparation, residual stress, porosity, inclusions, and loading configuration.

A 100 mm long carbide rod is therefore sensitive to grinding damage, edge chips, clamp pressure, bending, impact, thermal gradients, and resonance. The bulk self-stress calculation is favorable, but brittle-fracture reliability requires grade-specific Weibull statistics or proof testing, nondestructive inspection, controlled surface finish, and a containment enclosure.

The H10F rod geometry itself is commercially plausible: carbide-rod suppliers list 10 mm diameter and 100 mm length within normal blank ranges. Availability does not constitute a rotor certification.

14. Rotor dynamics and bending critical speed

The tensile calculations address axisymmetric centrifugal self-stress. They do not address bending whirl. For an ideal uniform simply-supported beam, the first bending natural frequency is

$$\omega_1 = \pi^2 \sqrt{E I_b / (\rho A L^4)}.$$

This estimate depends on stiffness-to-density ratio E/ρ and support spacing. Real bearing stiffness, overhang, end fixtures, motor coupling, gyroscopic effects, damping, and manufacturing runout can move the critical speed substantially.

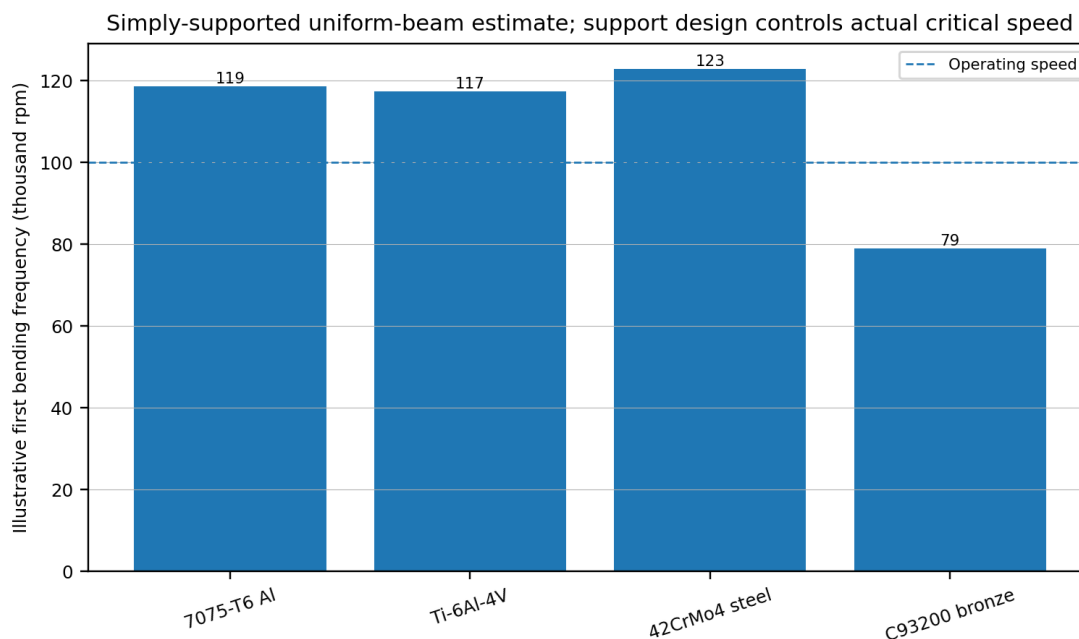


Figure 12. Illustrative simply-supported estimates for materials having a selected elastic modulus in the source ledger. Bronze is below 100,000 rpm in this simplified model, while aluminum, titanium, and steel are near the operating speed. This figure is a rotor-dynamic warning, not an operational rating.

Material	E GPa	Estimated first bending rpm	100k / estimate	Interpretation
7075-T6 Al	71.00	118,648	0.843	near estimate
Ti-6Al-4V	110.00	117,410	0.852	near estimate
42CrMo4 steel	210.00	122,889	0.814	near estimate
C93200 bronze	99.97	78,924	1.267	above estimate

A rotor can sometimes operate above a first critical speed, but it must pass through resonance during acceleration and deceleration. For an experimental apparatus, bearing placement should shorten the effective span, and measured modal testing should precede high-speed operation.

15. Material-by-material engineering assessment

WC-Co H10F produces the largest force, 17.062 mN, within the fixed envelope. Its bulk self-stress is small compared with typical TRS, but brittle flaw sensitivity, clamp design, grinding quality, and fragment containment dominate. It is the maximum-force material and the least forging of impact and local defects.

C93200 bronze produces 10.484 mN, more than the selected steel. Its conservative yield margin is the lowest, approximately 5.65. Its relatively low stiffness-to-density ratio also makes the 100 mm unsupported-span bending estimate unfavorable. Bronze is useful as a high-density comparison material but is not the preferred high-speed prototype rotor.

42CrMo4 / AISI 4140 quenched-and-tempered steel produces 9.084 mN. It combines high yield strength, high stiffness, toughness, machinability, inspectability, and a substantial conservative yield margin of approximately 42.5. It is the strongest general-purpose prototype candidate among the ductile materials.

Ti-6Al-4V Grade 5 produces 5.213 mN. Its high strength-to-density ratio gives an excellent conservative yield margin near 71, but its lower density limits force per fixed volume. It is well suited to high-speed verification where mechanical margin is prioritized over absolute force.

7075-T6/T651 aluminum produces 3.295 mN, the lowest force. It has an excellent strength-to-density ratio and the lowest bulk stress, but its lower stiffness and stress-corrosion sensitivities require careful environment, grain-direction, and fixture control. It is useful for low-energy commissioning and instrumentation checkout.

The fixed-envelope force ranking is controlled by density. The practical prototype ranking is different: 42CrMo4 steel offers the best balance of force, toughness, stiffness, machinability, and containment behavior; titanium and aluminum offer high strength-to-density; carbide offers maximum force but high brittleness; bronze offers density but comparatively low strength and stiffness.

16. Experimental and containment requirements

A 100,000 rpm test article is a high-energy rotor even when bulk tensile stress is modest. Each rotor must be individually characterized rather than accepted from nominal stock properties.

Requirement	Purpose
Dimensional metrology	Verify diameter, straightness, roundness, coaxiality, end geometry, and support span.
Dynamic balancing	Suppress synchronous bearing load and bending excitation.
Modal survey	Measure bending modes and support-dependent critical speeds before full-speed operation.
Overspeed proof test	Demonstrate margin in a remote, contained fixture.
Vacuum operation	Reduce aerodynamic heating and drag.
Bearing-temperature monitoring	Detect instability, preload error, and friction growth.
Radial displacement probes	Measure synchronous runout, subsynchronous whirl, and mode approach.
Six-axis force balance	Separate the predicted lateral signal from bearing and mount reactions.
Spin reversal	Reverse the K-Field force while preserving most static apparatus errors.
Geometric null orientation	Align the spin axis with g to test the exact force null.
Fragment containment	Retain ductile or brittle debris following a rotor failure.
Remote operation	Protect personnel during acceleration, overspeed, and coastdown.

The principal hazards are likely to be imbalance, bending resonance, bearing critical speed, anisotropy, fixture stress concentration, and fragment release rather than uniform centrifugal tensile stress. A calculated material safety factor is not an apparatus safety certification.

17. Conclusions

At the common 10 mm diameter, 100 mm length, optimum orientation, and 100,000 rpm, the predicted forces are 17.062 mN for H10F WC-Co carbide, 10.484 mN for C93200 bronze, 9.084 mN for 42CrMo4 steel, 5.213 mN for Ti-6Al-4V, and 3.295 mN for 7075-T6 aluminum.

Total force is exactly proportional to density for a fixed rotor envelope. Force per unit rotating mass is identical for all materials. Stored rotational energy also scales with mass, so force per stored energy is identical across the set.

The conservative thin-ring stress is 39.75 MPa for carbide, 24.43 MPa for bronze, 21.16 MPa for steel, 12.15 MPa for titanium, and 7.68 MPa for aluminum. These are deliberately harsher than an ideal solid-body stress solution.

The lowest ductile yield margin is bronze at approximately 5.65. Steel, aluminum, and titanium have large bulk-stress margins. The carbide TRS comparison is large but cannot be used directly as a design safety factor because the material is brittle and flaw-controlled.

For a first experimental rotor, quenched-and-tempered 42CrMo4 steel offers the strongest overall engineering balance. Ti-6Al-4V or 7075-T6 are useful for low-energy high-margin commissioning. H10F carbide maximizes force but requires brittle-fracture qualification and high-integrity containment. Bronze is a useful density control but is the least favorable ductile material at the stated slenderness and speed.

The determining mechanical question is not bulk centrifugal strength alone. The actual rotor must be designed around balance, bearing support, critical speeds, surface quality, attachment geometry, inspection, proof testing, and fragment containment.

Appendix A. Complete input and result ledger

Field	WC-Co H10F	7075-T6 Al	Ti-6Al-4V	42CrMo4 steel	C93200 bronze
Density kg/m ³	14500	2800	4430	7720	8910
Mass g	113.882733693	21.991148575	34.793138639	60.632738214	69.978976359
Force mN	17.062226406	3.294774754	5.212804343	9.084164680	10.484443950
Equivalent gf	1.739862890	0.335973524	0.531558110	0.926327001	1.069115748
Specific force N/kg	0.149822768148	0.149822768148	0.149822768148	0.149822768148	0.149822768148
Polar inertia kg m ²	1.423534171158e-06	2.748893571891e-07	4.349142329813e-07	7.579092276785e-07	8.747372044839e-07
Angular momentum kg m ² /s	1.490721498081e-02	2.878634616984e-03	4.554411197586e-03	7.936806872543e-03	9.160226584761e-03
Spin energy J	78.053995115	15.072495608	23.846841266	41.557023606	47.962834240
Ring stress MPa	39.752573282	7.676358979	12.145096527	21.164818327	24.427270893
Solid upper stress MPa	17.391750811	3.358407053	5.313479731	9.259608018	10.686931016
Yield SF ring	-	65.525856907	70.975146067	42.523398316	5.645329788
UTS SF ring	-	74.514493342	76.656451263	51.973042386	9.878303682
TRS ratio ring	113.200218966	-	-	-	-
Allowable rpm SF2	752330.443	572389.102	595714.470	461104.101	168007.884
Force/energy N/J	2.185951709567e-04	2.185951709567e-04	2.185951709567e-04	2.185951709567e-04	2.185951709567e-04

Appendix B. Complete 27-node centered Cell3 registry

#	2i	2j	2k	lambda	ICRS x	ICRS y	ICRS z
0	-1	-1	-1	-3	-0.010974902499	0.007385983128	0.039661911012
1	-1	-1	0	-2	-0.011290092242	-0.004742877881	0.018199733726
2	-1	-1	1	-1	-0.011605281985	-0.016871738890	-0.003262443560
3	-1	0	-1	-2	-0.009875529515	0.013837068257	0.022628660244
4	-1	0	0	-1	-0.010190719258	0.001708207248	0.001166482958
5	-1	0	1	0	-0.010505909001	-0.010420653761	-0.020295694328
6	-1	1	-1	-1	-0.008776156531	0.020288153386	0.005595409476
7	-1	1	0	0	-0.009091346274	0.008159292377	-0.015866767810
8	-1	1	1	1	-0.009406536017	-0.003969568632	-0.037328945096
9	0	-1	-1	-2	-0.000784183241	0.005677775880	0.038495428054
10	0	-1	0	-1	-0.001099372984	-0.006451085129	0.017033250768
11	0	-1	1	0	-0.001414562727	-0.018579946138	-0.004428926518
12	0	0	-1	-1	0.000315189743	0.012128861009	0.021462177286
13	0	0	0	0	0.000000000000	0.000000000000	0.000000000000
14	0	0	1	1	-0.000315189743	-0.012128861009	-0.021462177286
15	0	1	-1	0	0.001414562727	0.018579946138	0.004428926518
16	0	1	0	1	0.001099372984	0.006451085129	-0.017033250768
17	0	1	1	2	0.000784183241	-0.005677775880	-0.038495428054
18	1	-1	-1	-1	0.009406536017	0.003969568632	0.037328945096
19	1	-1	0	0	0.009091346274	-0.008159292377	0.015866767810
20	1	-1	1	1	0.008776156531	-0.020288153386	-0.005595409476
21	1	0	-1	0	0.010505909001	0.010420653761	0.020295694328
22	1	0	0	1	0.010190719258	-0.001708207248	-0.001166482958
23	1	0	1	2	0.009875529515	-0.013837068257	-0.022628660244
24	1	1	-1	1	0.011605281985	0.016871738890	0.003262443560
25	1	1	0	2	0.011290092242	0.004742877881	-0.018199733726
26	1	1	1	3	0.010974902499	-0.007385983128	-0.039661911012

Appendix C. Cell3 vertices and face families

#	i	j	k	ICRS x	ICRS y	ICRS z
0	-0.5	-0.5	-0.5	-0.010974902499	0.007385983128	0.039661911012
1	-0.5	-0.5	0.5	-0.011605281985	-0.016871738890	-0.003262443560
2	-0.5	0.5	-0.5	-0.008776156531	0.020288153386	0.005595409476
3	-0.5	0.5	0.5	-0.009406536017	-0.003969568632	-0.037328945096
4	0.5	-0.5	-0.5	0.009406536017	0.003969568632	0.037328945096
5	0.5	-0.5	0.5	0.008776156531	-0.020288153386	-0.005595409476
6	0.5	0.5	-0.5	0.011605281985	0.016871738890	0.003262443560
7	0.5	0.5	0.5	0.010974902499	-0.007385983128	-0.039661911012

Family	Area native^2	Included angle	Unit normal ICRS
A_ab	7.547217896293e-04	83.933483014 deg	(0.1940922008, 0.9131771688, 0.3583792348)
A_ac	1.011255757668e-03	80.448130021 deg	(0.0890525941, 0.8665767693, -0.4910339483)
A_bc	1.385784431744e-03	50.363231813 deg	(-0.9959651903, 0.0836021625, -0.0326192906)

Basis-vector details

Vector	ICRS x	ICRS y	ICRS z	Length
a	0.020381438516	-0.003416414496	-0.002332965916	0.020797058781
b	0.002198745968	0.012902170258	-0.034066501536	0.036494205130
c	-0.000630379486	-0.024257722018	-0.042924354572	0.049308565901

Appendix D. Source ledger

ID	Organization	Document	Data used
S1	Hyperion Materials & Technologies	Insert Blanks Grade Properties	H10F density 14.5 g/cm ³ ; typical TRS 4500 MPa; compressive strength 6300 MPa
S2	Kaiser Aluminum	Rod & Bar Alloy 7075 Technical Data	T6/T651 UTS 572 MPa; yield 503 MPa; modulus 71.0 GPa; nominal density 2.80 Mg/m ³
S3	Aircraft Materials	Titanium Alloy Grade 5 / Ti-6Al-4V Technical Data Sheet	Annealed bar below 2 in: minimum UTS 931 MPa; yield 862 MPa
S4	TIMET	TIMETAL 6-4 alloy data family	Grade identity and conservative cross-check; density near 4.42-4.43 g/cm ³ ; elastic modulus representative value
S5	Swiss Steel Group	1.7225 / 42CrMo4 Material Data Sheet	Density 7.72 kg/dm ³ ; E=210 GPa; for d<=16 mm yield >=900 MPa and UTS 1100-1300 MPa
S6	ASTM International	ASTM B406-96(2025), Transverse Rupture Strength of Cemented Carbides	TRS is statistically flaw-sensitive and is not a design value
S7	Copper Development Association	C93200 Alloy Data	Specific gravity 8.91; continuous-cast UTS 35 ksi and yield 20 ksi; tensile modulus 14500 ksi

Source locations

S1: <https://www.hyperionmt.com/en-vn/products/Insert-Blanks/grade-properties/>

S2: https://online.kaiseraluminum.com/depot/PublicProductInformation/Document/1028/Kaiser_Aluminum_7075_Rod_and_Bar.pdf

S3: <https://www.aircraftmaterials.com/data/titanium/ti6al4v.html>

S4: <https://www.timet.com/documents/datasheets/alpha-and-beta-alloys/>

S5: <https://swisssteel-group.com/en/services/download-center/1-7225-1-7227-42crmo4-42cromo4-werkstoffdatenblatt>

S6: <https://store.astm.org/b0406-96r25.html>

S7: <https://alloys.copper.org/alloy/C93200>

The source property values are representative of the named grade and condition. Final rotor design requires certified lot data, product-form verification, and manufacturer acceptance for the intended rotating application.

Appendix E. Compact equations and reconstruction

Relationship	Equation
Rod volume	$V = \pi R^2 L$
Mass	$m = \rho V$
Angular speed	$\omega = 2 \pi \text{ RPM}/60$
Rim speed	$v = \omega R$
Field response	$g = G_{\text{phase}} V_E$
Force	$F = (\kappa_F m \omega/2) \hat{n} \times g$
Optimum force	$F_{\text{max}} = m (\kappa_F g /2) \omega$
Specific force	$F_{\text{max}}/m = (\kappa_F g /2) \omega$
Thin-ring stress bound	$\sigma_{\text{ring}} = \rho v^2$
Ideal solid center stress	$\sigma_{\text{center}} = ((3+\nu)/8) \rho \omega^2 R^2$
Poisson-independent solid upper bound	$\sigma_{\text{center}} \leq (7/16) \rho v^2$
Polar inertia	$I = (1/2) m R^2$
Angular momentum	$H = I \omega$
Spin energy	$E = (1/2) I \omega^2$
Force per energy	$F/E = 4 q_g/(R^2 \omega)$
Strength-limited rpm	$\text{RPM} = [60/(2 \pi R)] \sqrt{(S/(N \rho))}$
Simply-supported bending frequency	$\omega_1 = \pi^2 \sqrt{(E I_b/(\rho A L^4))}$

Minimal reconstruction sequence

1. Load the rod dimensions, rpm, material density, and selected strength data.
2. Compute volume, mass, angular speed, rim speed, polar inertia, angular momentum, and spin energy.
3. Load G_{phase} , V_E , c_{hat} , and κ_F ; compute g , n_{opt} , and F_{hat} .
4. Compute the common specific force $q_g \omega$ and multiply by each material mass.
5. Compute the thin-ring conservative stress and the ideal-solid upper stress.
6. Divide yield, UTS, or TRS by the selected stress model.
7. Solve the ring-bound strength equation for allowable speed at the required ratio.
8. Perform a separate support-dependent rotor-dynamic analysis before operation.

Appendix F. Archival record

Field	Record
Document title	K-Field Rod Material Comparison
Principal Investigator	P. Dwayne Esterline
Education	BS, Huntington University (1999); MBA, Indiana Wesleyan University (2008)
Organization	Krytronx
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.
Generation UTC	2026-07-26T19:12:57Z
Generation epoch	1785093177
Output PDF	K-Field_rod_material_comparison_1785093177.pdf
Machine-readable companion	K-Field_rod_material_comparison_1785093177.json
Report generation	Python ReportLab; built-in core fonts only
Figure generation	Python numerical computation and raster rendering from the exact report ledger
Coordinate frame	ICRS/J2000 Cartesian
Mechanical units	SI
Cell3 units	native source units
Source format references	K-Field tubular rotor and solid-rod reports; manufacturer material data; ASTM carbide test standard

P. Dwayne Esterline

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